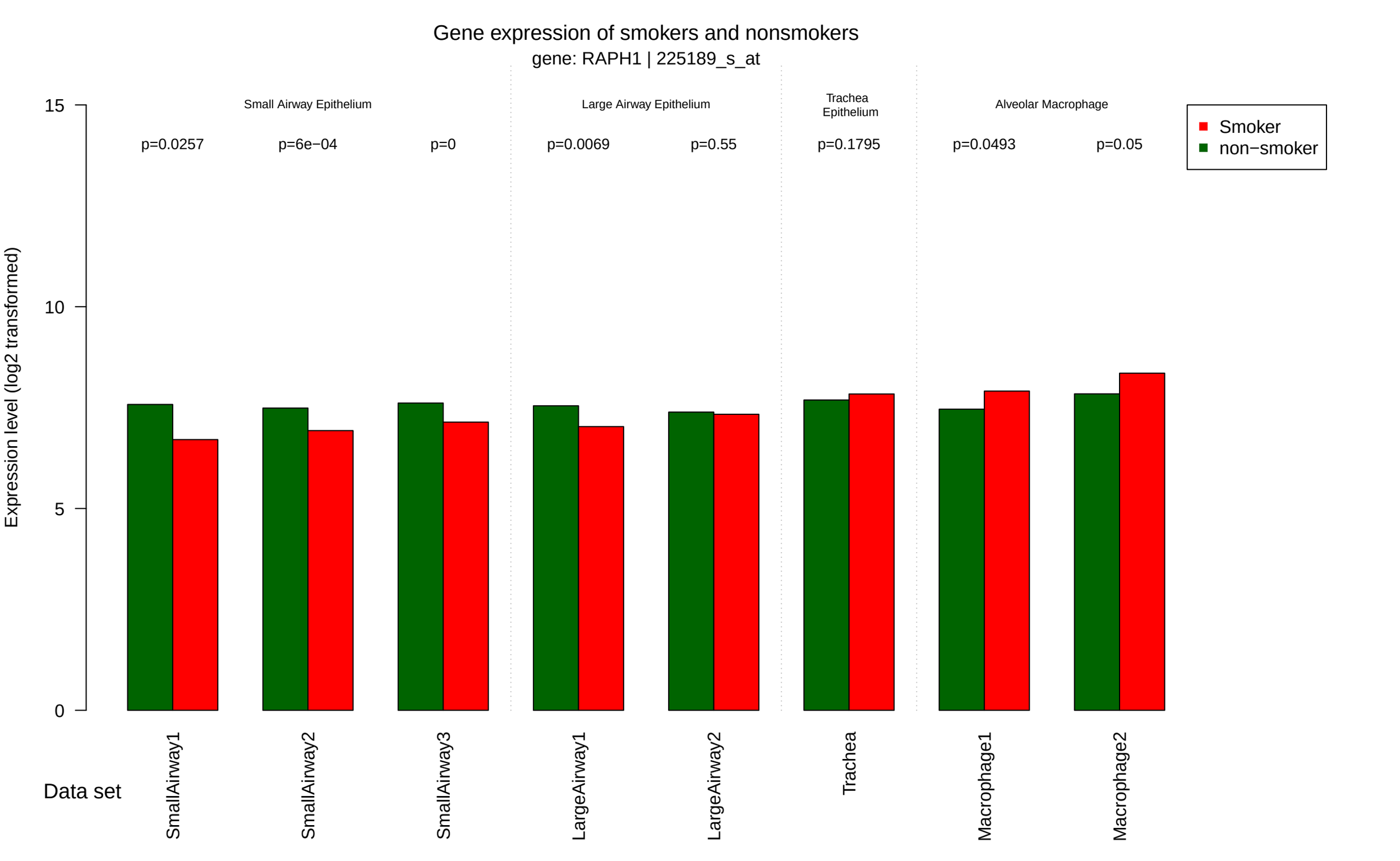
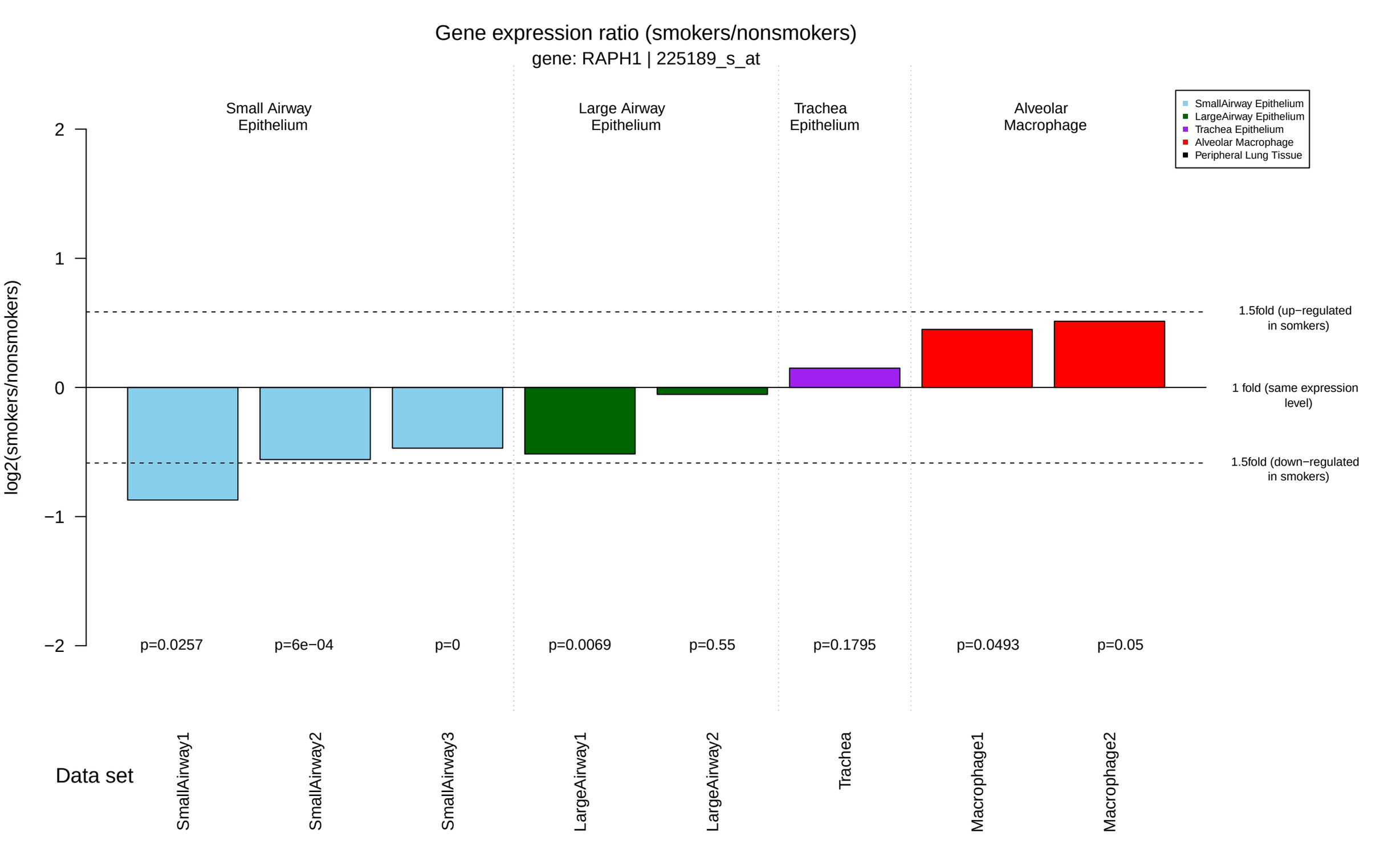
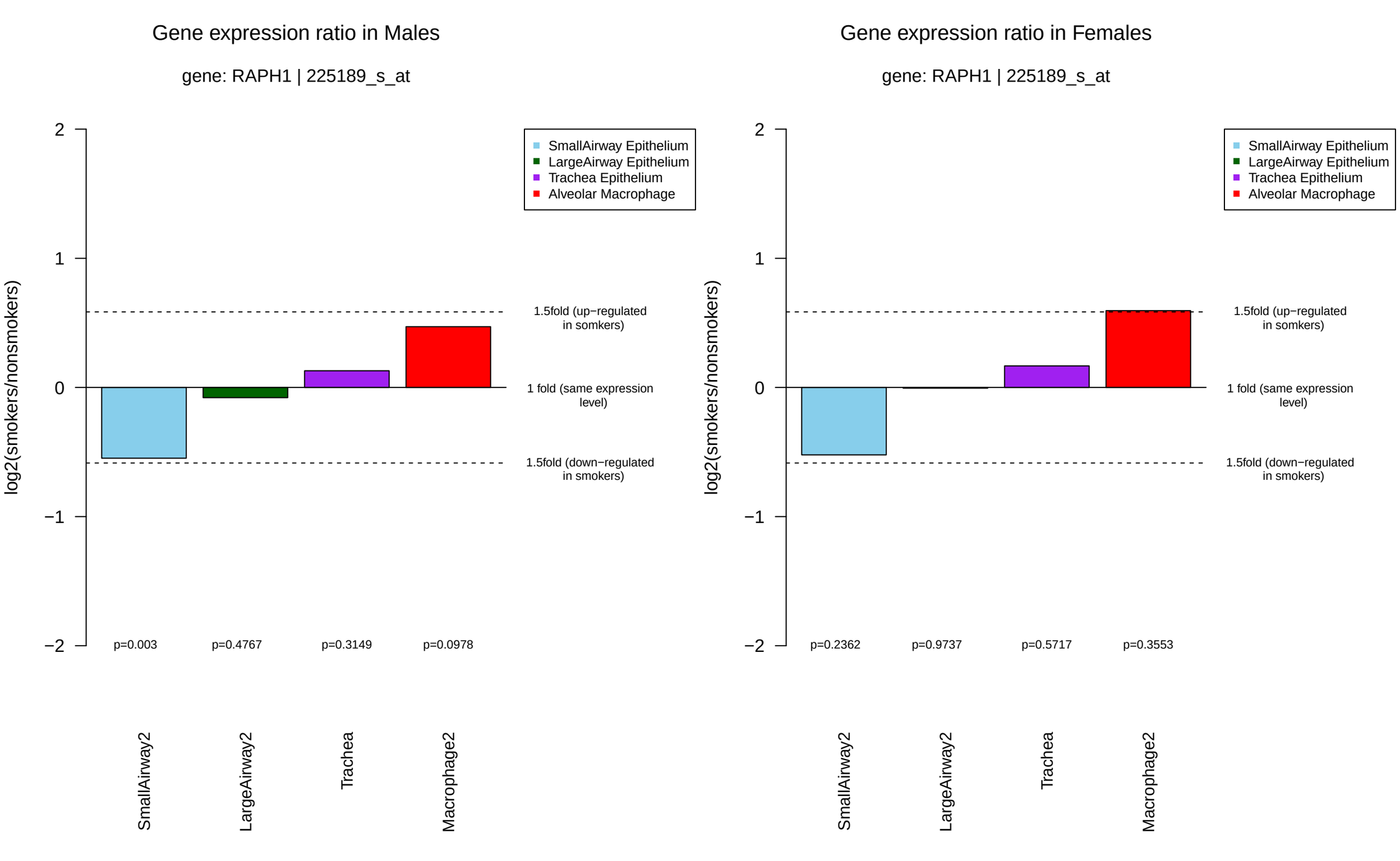
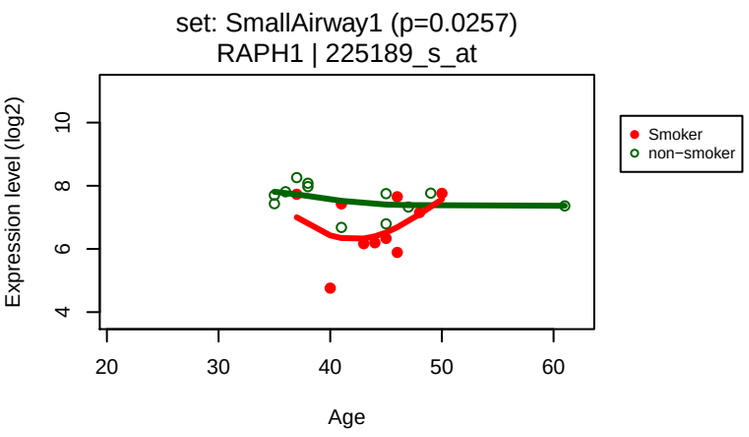
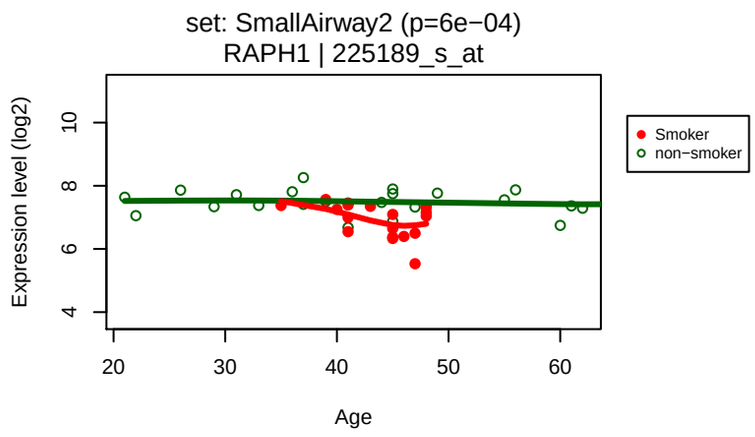
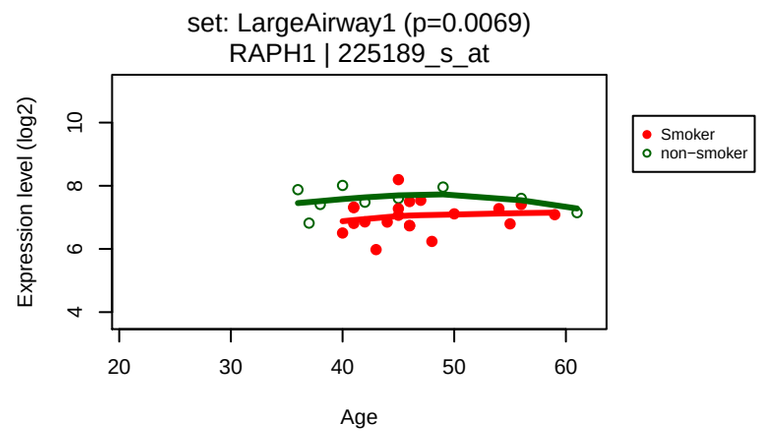

Gene expression of smokers and nonsmokers
gene: RAPH1 | 225189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0257
p=6e−04
p=0
p=0.0069
p=0.55
p=0.1795
p=0.0493
p=0.05
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

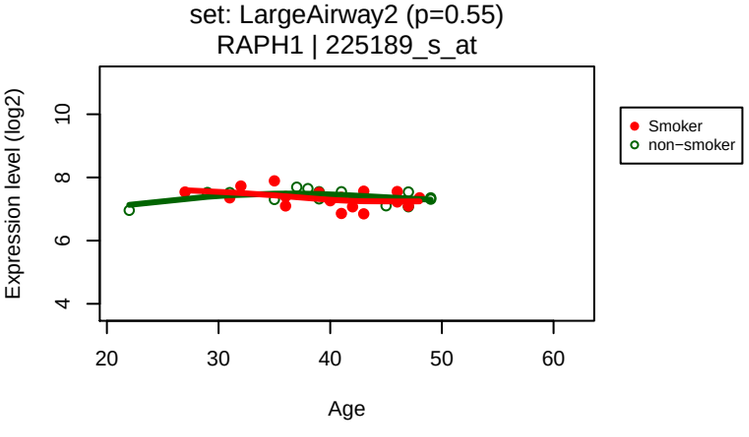

Gene expression ratio (smokers/nonsmokers)
gene: RAPH1 | 225189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=6e−04
p=0
p=0.0069
p=0.55
p=0.1795
p=0.0493
p=0.05
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

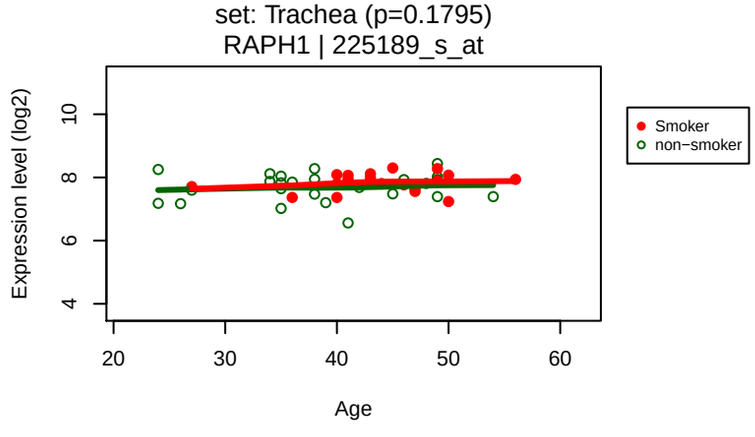

Gene expression ratio in Males
gene: RAPH1 | 225189_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.003
p=0.4767
p=0.3149
p=0.0978
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPH1 | 225189_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2362
p=0.9737
p=0.5717
p=0.3553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

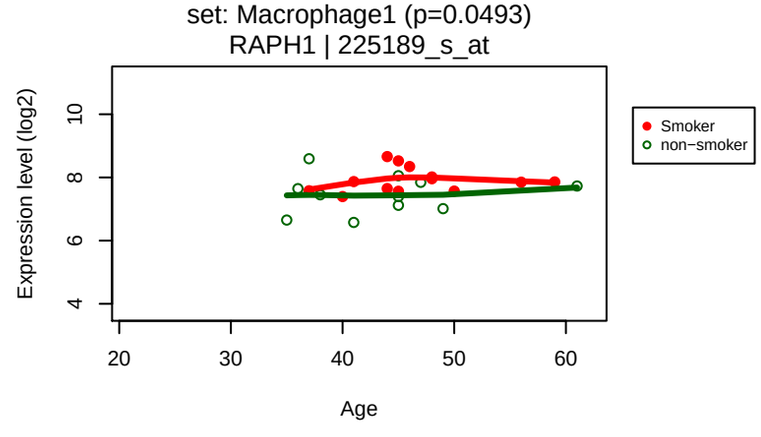

set: SmallAirway1 (p=0.0257)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker

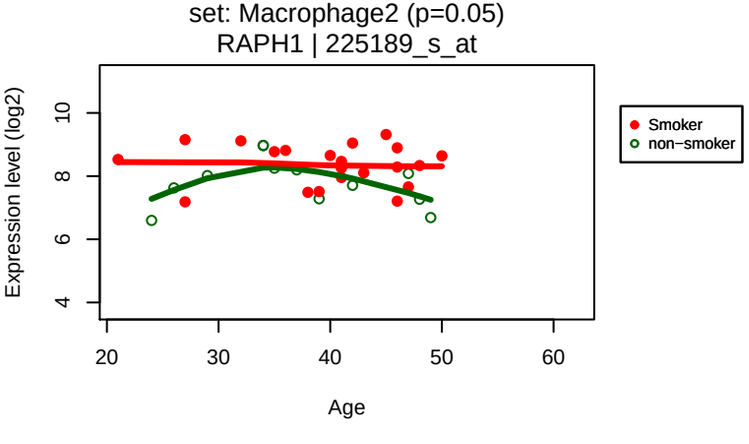

set: SmallAirway2 (p=6e−04)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker

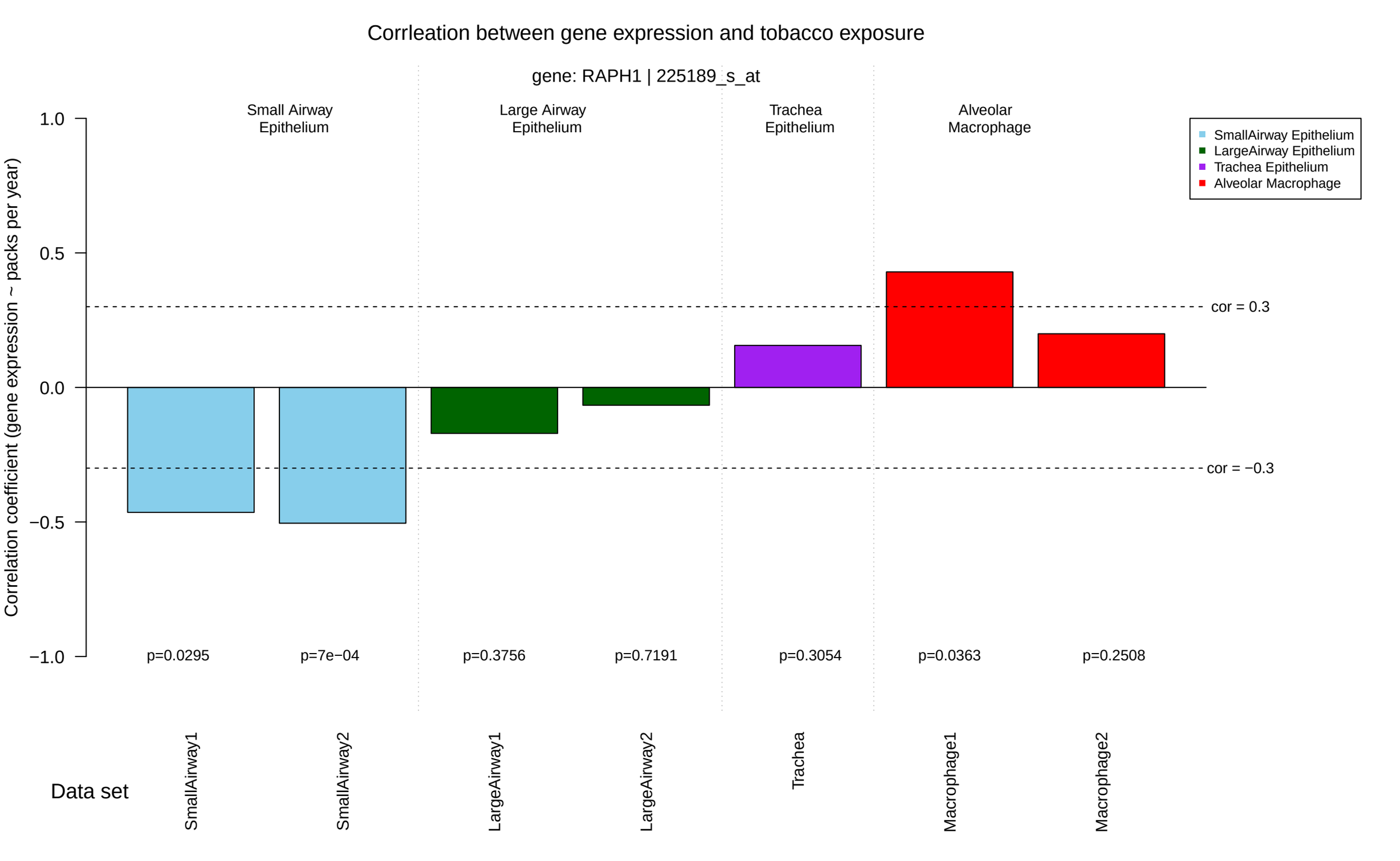

set: LargeAirway1 (p=0.0069)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.55)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1795)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0493)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.05)
RAPH1 | 225189_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPH1 | 225189_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0295
p=7e−04
p=0.3756
p=0.7191
p=0.3054
p=0.0363
p=0.2508
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2